Example 4.33 In ΔABC, points D, E, F lies on BC, CA, AB respectively. Suppose AB, AC and BC have lengths 13, 14 and 15 respectively. If $\frac{AF}{FB} = \frac{2}{5}$ and $\frac{CE}{EA} = \frac{5}{8}$. Find BD and DC.

Solution Given that $AB = 13$, $AC = 14$ and $BC = 15$.

Let $BD = x$ and $DC = y$

Using Ceva's theorem, we have, $\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1$ ...(1)

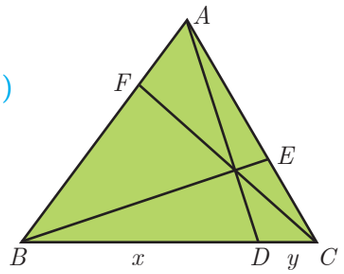


Fig. 4.73

Substitute the values of $\frac{AF}{FB}$ and $\frac{CE}{EA}$ in (1),

we have $\frac{BD}{DC} \times \frac{5}{8} \times \frac{2}{5} = 1$

$\frac{x}{y} \times \frac{10}{40} = 1$ we get, $\frac{x}{y} \times \frac{1}{4} = 1$. Hence, $x = 4y$ ...(2)

$BC = BD + DC = 15$ so, $x + y = 15$ ... (3)

From (2), using $x = 4y$ in (3) we get, $4y + y = 15$ gives $5y = 15$ then $y = 3$

Substitute $y = 3$ in (3) we get, $x = 12$. Hence $BD = 12$, $DC = 3$.

Example 4.34 In a garden containing several trees, three particular trees P, Q, R are located in the following way, $BP = 2$ m, $CQ = 3$ m, $RA = 10$ m, $PC = 6$ m, $QA = 5$ m, $RB = 2$ m, where A, B, C are points such that P lies on BC, Q lies on AC and R lies on AB. Check whether the trees P, Q, R lie on a same straight line.

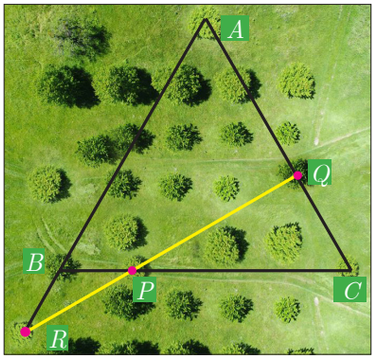


Fig. 4.74

Solution By Meanlau's theorem, the trees P, Q, R will be collinear (lie on same straight line)

if $\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{RA}{RB} = 1$ ...(1)

Given $BP = 2$ m, $CQ = 3$ m, $RA = 10$ m, $PC = 6$ m, $QA = 5$ m and $RB = 2$ m

Substituting these values in (1) we get, $\frac{BP}{PC} \times \frac{CQ}{QA} \times \frac{RA}{RB} = \frac{2}{6} \times \frac{3}{5} \times \frac{10}{2} = \frac{60}{60} = 1$

Hence the trees P, Q, R lie on a same straight line.

Progress Check

1. A straight line that touches a circle at a common point is called a _____.
2. A chord is a subsection of _____.
3. The lengths of the two tangents drawn from _____ point to a circle are equal.
4. No tangent can be drawn from _____ of the circle.
5. _____ is a cevian that divides the angle, into two equal halves.

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